

Verity — Validation Report

Hamby (2009) Barrel Set 252

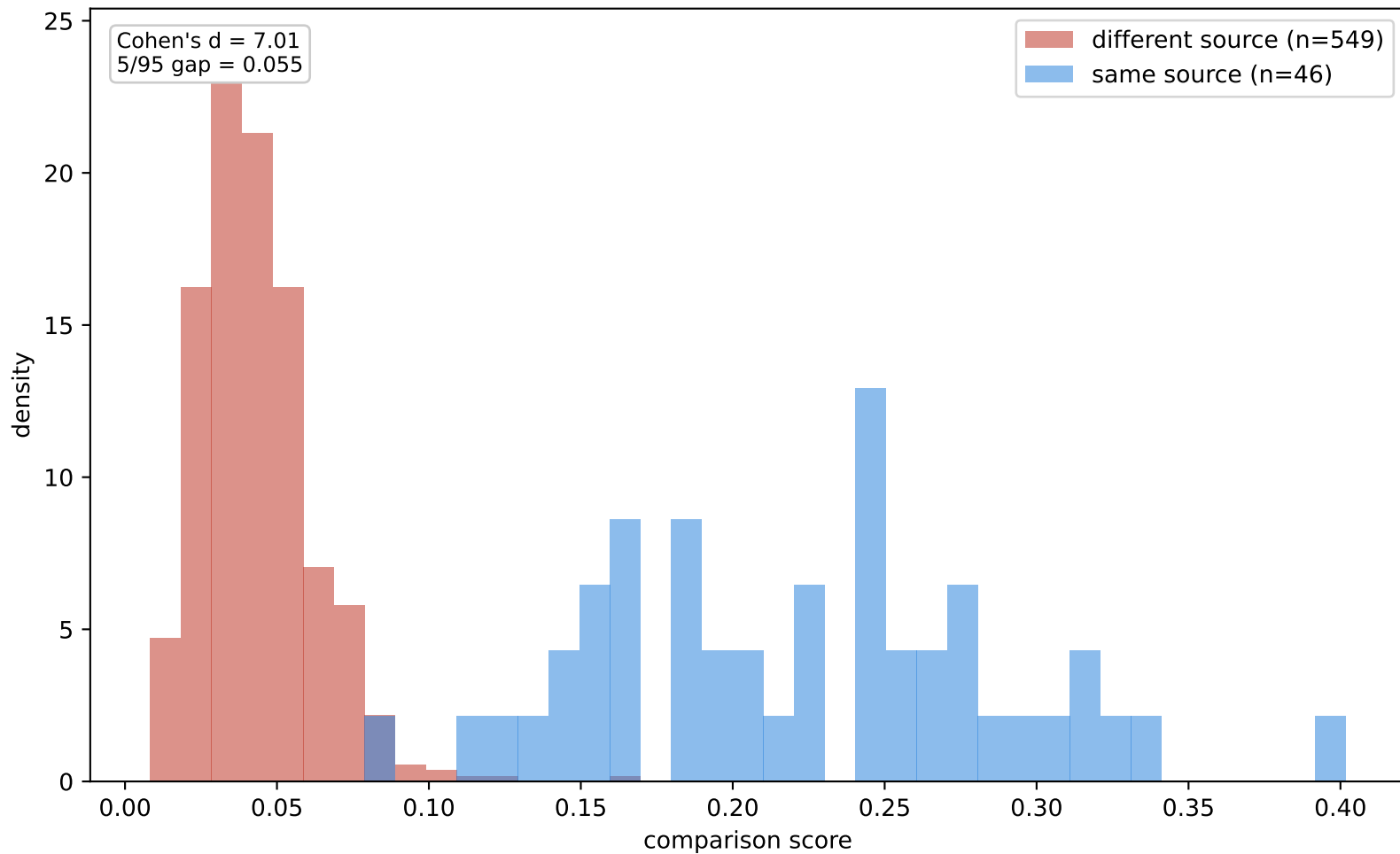
Domain	striated
Same-source comparisons (KM)	46
Different-source comparisons (KNM)	549
Distinct sources (barrels)	10
Pipeline version	0.1.0
Generated	—

Headline

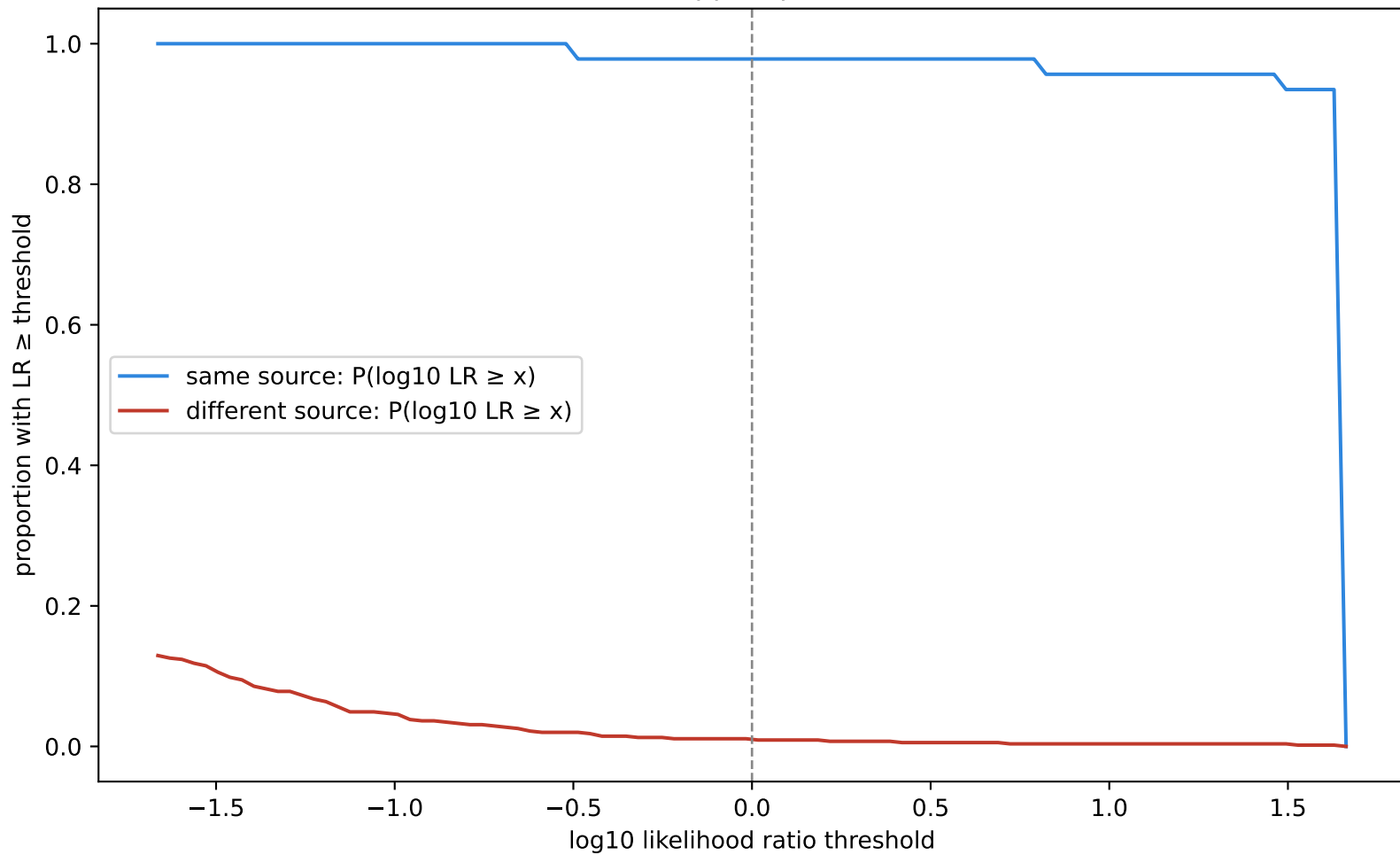
AUC = 0.999 Cllr = 0.077 Cllr_min = 0.032

Scope. This report characterizes the weight of evidence on the named reference population above; it is not a claim about the error rate of striated examination, which remains unknown. The likelihood ratio is calibrated by a monotone, bounded (ELUB) transform of the comparison score and is reported with a characterized cost (Cllr). Out-of-sample behaviour is summarized on the source-disjoint page.

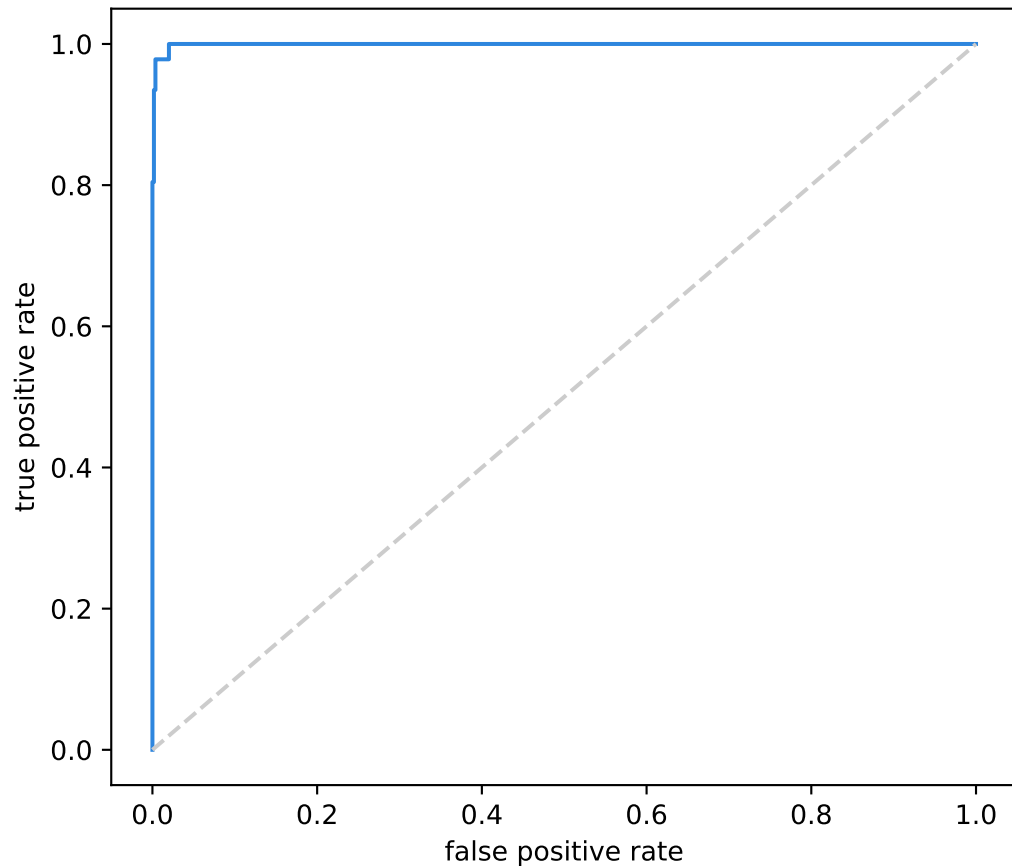
Score distributions — same vs different source



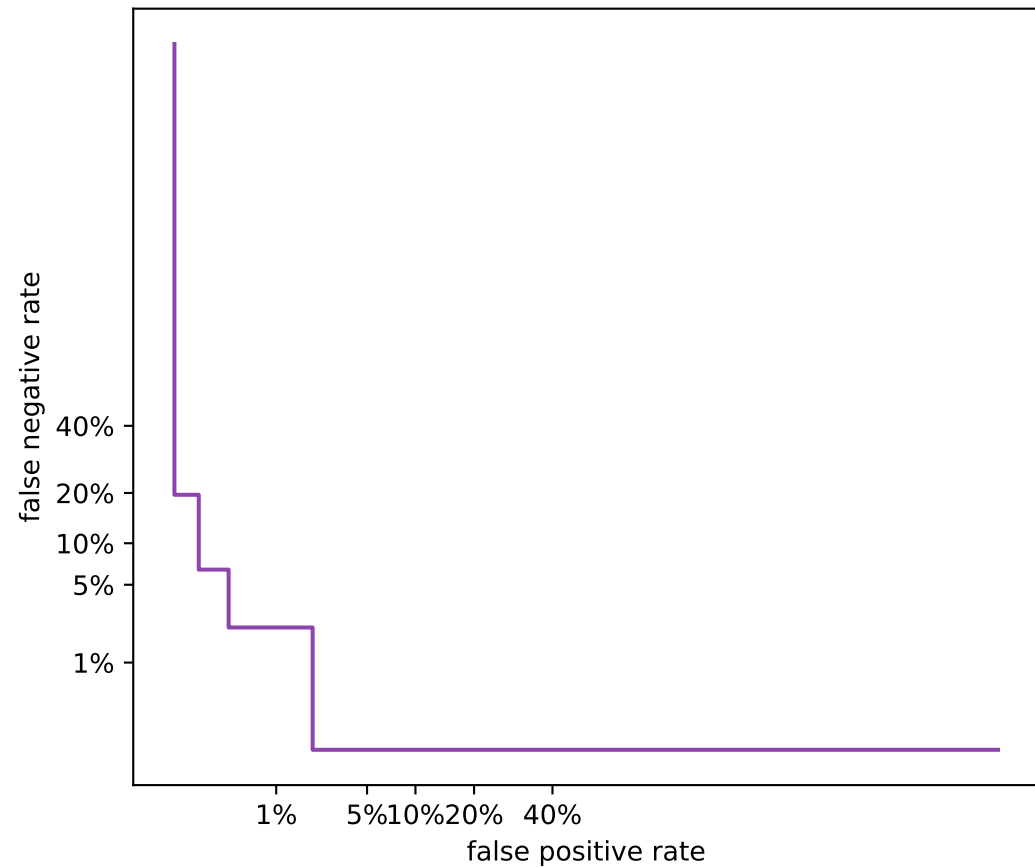
Tippett plot



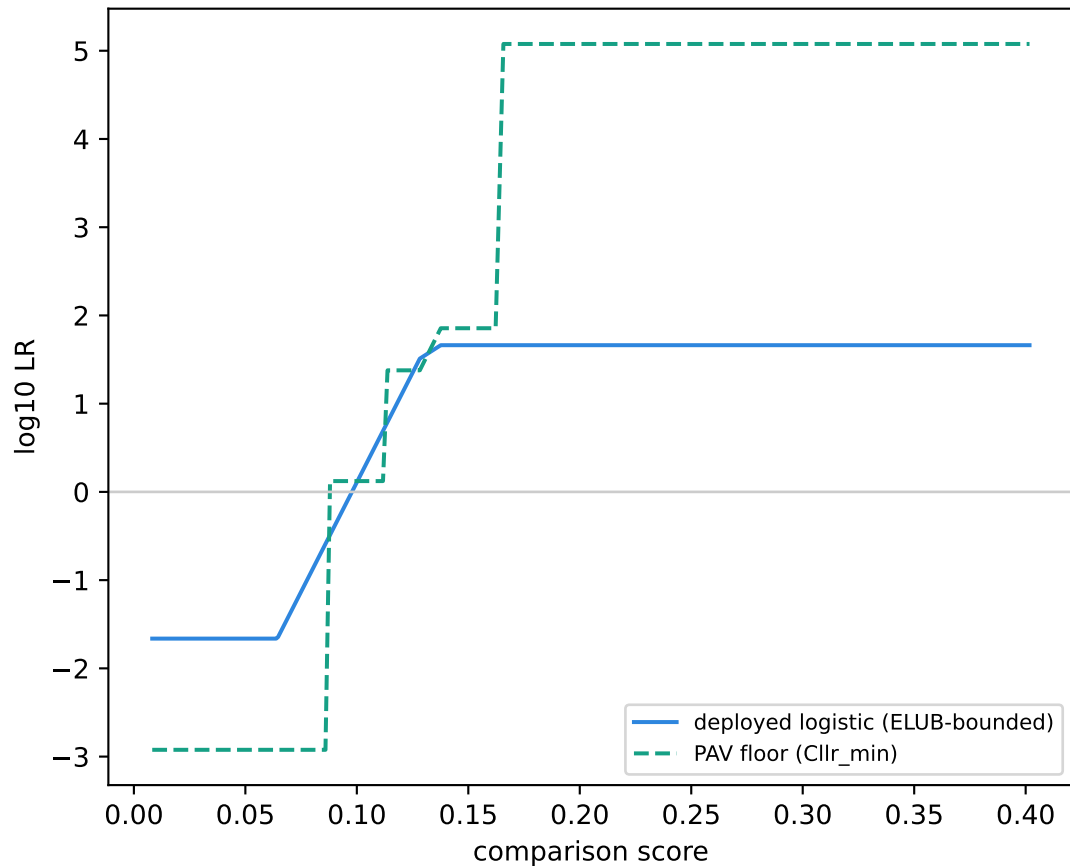
ROC (AUC = 0.999)



DET (EER = 2.1%)



Calibration map (monotone — the audit firewall)



Calibration decomposition

Cllr (deployed)	0.077
Cllr_min (floor)	0.032
calibration loss	+0.044
ECE	0.031
AUC	0.999
EER	2.1%
ELUB bound $ \log_{10} LR $	1.66

Source-disjoint validation

No barrel spans train and test. Mean $\pm$ sd over 10 splits — the out-of-sample weight-of-evidence cost. This is the characterization that answers Cuellar et al. (2024): a Cllr on a named dataset under a source-disjoint protocol.

test Cllr	0.113 $\pm$ 0.066
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test Cllr_min	0.004 $\pm$ 0.011
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test AUC	1.000 $\pm$ 0.001
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calibration loss	+0.109
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